

Problem Statement

Emotional-support chatbots are built for English, but many Indians share distress in Roman-script Hinglish (“*yaar result aaya, bahut tension ho rahi hai, ghar pe kaise bataun*”).

MultiAgentESC (EMNLP 2025), a strong LLM multi-agent ESC system, (i) retrieves past cases with an English-only encoder, (ii) picks replies without looking at the user’s language mix, and (iii) was never tested on code-mixed users. Goal: make it work for Hinglish help-seekers without fine-tuning the LLM agents.

1) Group Name: CodeMixESC (individual project) **2) Team Name, Roll no:** Roshan Raj — 12341830

3) Novelty

1 Code-Mix Profiler + Register Gate

HingBERT-LID + Code-Mixing Index measure how much Hindi the user mixes; the gate re-writes a reply whose mix is off (max. 1 extra call).

2 Cross-lingual retriever

Multilingual mpnet fine-tuned contrastively on 16,951 Hinglish–English pairs: Hinglish queries find the right English cases.

3 Register-aware selection

Agents see the user’s register; debate, voting and refiner add a 4th criterion: language & cultural fit.

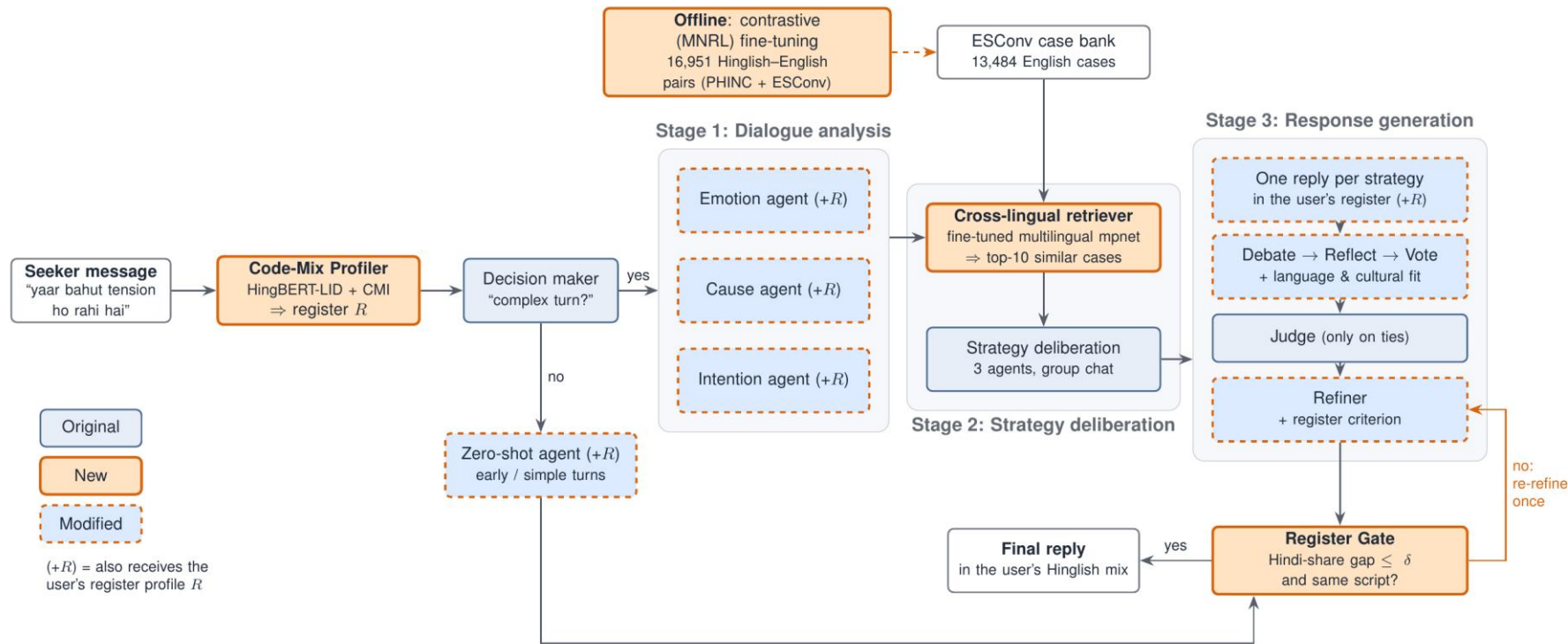
4 ESConv-HiEn benchmark

First Hinglish ESC test set with strategy labels: ESConv test set rewritten at Light and Heavy mixing, CMI-verified.

4) Contributions

Roshan Raj (sole member, 100%): designed the method; built ESConv-HiEn (LLM rewriting + CMI and meaning checks); built the 16,951 training pairs and fine-tuned the retriever; re-implemented MultiAgentESC with all four components and 4 baselines (zero-shot, few-shot CoT, MultiAgentESC, translate-pivot) + ablations; built the evaluation (BLEU, ROUGE-L, chrF, BERTScore, CMI gap, strategy JSD, LLM judge, conversation-level bootstrap) with 110 unit tests and CI.

Architecture: CodeMixESC



Results and Findings

A. Retrieval robustness (%)

Retriever (test, 1,210 turns)	Overlap Light	Overlap Heavy	P@10 Light	P@10 Heavy
RoBERTa-L (base paper)	56.9	27.0	35.8	33.8
LaBSE	61.4	33.2	28.9	28.5
Multiling. mpnet	69.9	31.3	31.6	28.4
mpnet fine-tuned (ours)	73.2	53.3	32.0	31.5

B. ESConv-HiEn (new dataset)

ESConv test set	Conv.	Utterances	Hindi words	Mean CMI	In target band	Sim. to English
English original	100	3,140	1%	0.01	—	—
Light Hinglish (CMI 0.1-0.3)	100	3,140	16%	0.16	94.5%	0.88
Heavy Hinglish (CMI 0.3-0.5)	100	3,140	50%	0.34	87.5%	0.71

C. Reply quality (26B agents, same 50 turns)

System (50 turns)	R-L Light	R-L Heavy	CMI gap Light	CMI gap Heavy	Calls
MultiAgentESC	10.45	5.66	0.141	0.357	10.2
Translate-Pivot	5.75	8.62	0.136	0.208	11.2
CodeMixESC (ours)	10.60	9.24	0.137	0.120	10.7

Key findings

- Heavy code-mixing breaks the English retriever: only **27%** of its top-10 cases match the English query's; our fine-tuned retriever keeps **53%** (+26.3 points, 95% CI [24.6, 28.2]) and the retrieved strategies stay stable (JSD 0.15 $\rightarrow$ 0.01); trade-off: problem-type precision -2 to -4 points.
- Replies (26B agents): Heavy vs MultiAgentESC: ROUGE-L +3.58 [+1.83, +5.27], $p < 0.01$; Light vs MultiAgentESC: ROUGE-L +0.15 [-1.75, +2.09], $p = 0.93$; Register Gate CMI gap -0.017 [-0.033, -0.001], $p = 0.04$.
- LLM judge (gemma-4-31b, both orders): CodeMixESC preferred over MultiAgentESC overall in **86% vs 4%** of Heavy turns and **74% vs 20%** of Light turns (win vs loss, $p < 0.01$).

► **Live demo:** codemixesc.streamlit.app **Code, data & results:** github.com/roshanraj9136/CodeMixESC